

Introduction to Machine Learning

Machine learning is a branch of artificial intelligence. Systems learn from data instead of following explicit rules. There are three main types: supervised learning, unsupervised learning, and reinforcement learning.

What is an Embedding

An embedding is a vector of numbers that encodes the meaning of text. Semantically similar texts have similar vectors. The distance between vectors is measured using cosine similarity.

What is RAG

RAG stands for Retrieval-Augmented Generation. It combines search over a knowledge base with text generation by a language model. Documents are split into chunks, embedded, and stored in a vector database. At query time the most relevant chunks are retrieved and passed to the LLM.

Context Window and Tokens

A token is the basic unit of text processed by a language model. One token is roughly 4 characters in English. The context window is the maximum number of tokens a model can process at once.